



UCD College of Business

UCD College of Business, Global Campus

Assessment Submission Form

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Assessment Title:	BBS41 Sg FT DADM Assignment
Module Code:	MIS3008S
Module Title:	Data Analysis for Decision Makers

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Signed:

1.Han Bin	2.Chong Shi Xian
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Date: 17 June 2025

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BBS41 Sg FT DADM Assignment

Data Visualization and Analysis Report of Adidas US Sales

Understanding the Key Drivers of Adidas US Sales Growth and
Scaling Them for Broader Market Success

Group 5

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Word Count: 2698 words

Executive Summary

Objective of the report

This report aims to find out Adidas US existing most effective business strategy drive the growth, duplicate, and implement across region to increase overall sales performance and profitability of Adidas US.

Method

In this report we use data visualisation to show the sales trend overtime to identify the outstanding growth. The combination of statistical table and YOY growth rate calculation formula being used to shows clear comparison between dimension and amount increased.

We also make use of colour-coded representation to shows proportion comparison. Apart from that, we utilize drill-down analysis to explore the root driver of growth. Break down and segmented analysis both frequently been used in this report.

After identifying the significant data point, we hence segmented by season, by states and the like. Segmented data will be used to compare and identify the relative value differences, trend difference, and relative magnitude across various groups.

Result

2021 sales growth by 621.37% year over year. Online sales result 2021 explosively grew 7425.08% year over year, shows biggest growth and achieve highest 45.50% profit margin and largest proportion of total sales among all sales methods.

California, Florida, New York is the top 3 highest sales states in 2021. Online sales grew 19097%, 4627%, 3411% year-over-year. Online sales driven the growth, proving that the strategy of focusing on online sales is successful.

Top 3 state has significant performance lift in November and December differ from other state. We assume that top 3 state strategically boost sales in peak online sales season (November and December).

Dashboard 5 show if other state implements the same strategy and achieve same performance lift, total sales will growth 9.44%.

Conclusion (Business Implication)

Online sales have widely promoted, only small number of states catching the peak season to achieve significant growth, if the rest can replicate the effective strategy, they will be 9.445 growth in overall performance improve Adidas US overall sales performance.

Business Understanding

Intro of company

Adidas, full name is Adidas AG, properly founded in 1949 by Adolf Dassler. Adidas head quarter in Bavaria, Germany. Adidas is now the second largest sportwear manufacture company in the world, as well as the biggest sportwear manufacturer in Europe.

Business problem

Adidas US 2021 achieve ideal profit margin and sales explosively growth, 621.37% year-over-year growth in 2021 compared to 2020. However, which is not a uniform growth across all regions, many state doesn't catching the growth opportunity, miss the opportunity to maximize sales during peak online sales seasons which show the problem of lack of proper consistent implementation of effective strategies across the whole market.

Business objective

The business objective of this report is defining the key driver of 2021 growth. Through data analysis to identify the strategy most effectively boost the sales performance hence, replicating and scale up strategic action to weaker segments to achieve 9% sales growth in overall sales.

Datamining objective

The datamining objective is to analyse Adidas US sales record to find out what is the outstanding performance patterns and compare with underperforming segments, to find out the key driver behind the success, then identify the most effective proven strategy.

In this report we intent to use data visualisation, analyse geographic sales data, monthly sales data, different sales methods performance, sales method profitability, sales contribution by channel, sales growth rate and the like. To get a understanding of strategy, then identify which strategy most effectively boost the sales performance. The ultimate objective is to provide replicable and scalable strategic recommendations that help Adidas US improve overall sales performance and profitability through data-driven decision-making.

Modal

Profit = Sales x Profit Margin.

Data Understanding

Brief Overview

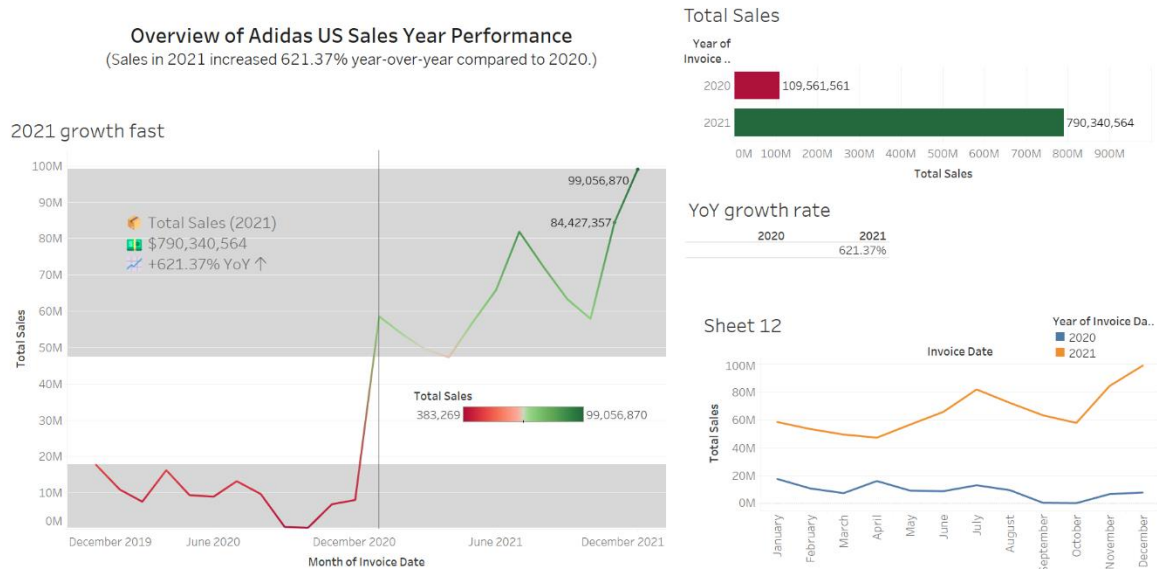
This document focuses on online sales data from 2020 to 2021. It highlights sales trends across U.S. states, especially in California, New York, and Florida—the top performers. The analysis examines how online sales changed over time, with special attention to seasonal peaks in November and December. It also explores how successful strategies from the top states could be applied elsewhere. The data presents both total and average monthly online sales by state. It includes profit margin comparisons between different sales channels: online, in-store, and outlet. Online sales had the highest average profit margin at 45.5%. The charts emphasize the sharp rise in online sales in 2021, which grew by 7425% year-over-year (YoY). This growth played a major role in boosting total revenue and profit.

Various groups of variables

This document focuses on online sales data from 2020 to 2021. It highlights sales trends across U.S. states, especially in California, New York, and Florida—the top performers. The analysis examines how online sales changed over time, with special attention to seasonal peaks in November and December. It also explores how successful strategies from the top states could be applied elsewhere. The data presents both total and average monthly online sales by state. It includes profit margin comparisons between different sales channels: online, in-store, and outlet. Online sales had the highest average profit margin at 45.5%. The charts emphasize the sharp rise in online sales in 2021, which grew by 742.5% year-over-year (YoY). This growth played a major role in boosting total revenue and profit.

Business Insights

Dashboard 1



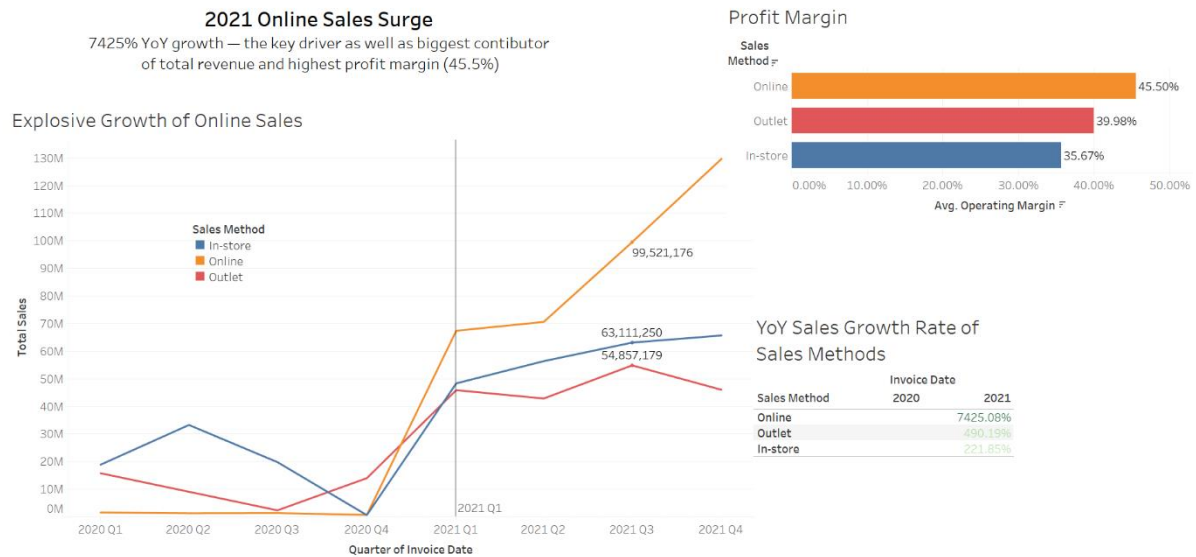
Dashboard 1 presents Adidas' sales performance in the U.S. market for 2021. It highlights a major rebound from the previous year. Compared to 2020, Adidas saw a dramatic increase in sales—more than seven times higher. This signals a strong recovery and rapid growth following the challenges of the pandemic. In 2021, Adidas reached total annual sales of \$790,340,564 in the U.S. This was a 621.37% increase over 2020. The sharp rise marked a key turning point for the brand. Monthly sales data, shown in the lower charts, tell a clear story. In 2020, many months reported online sales below \$10 million. The company struggled throughout the year.

However, beginning in January 2021, sales began to climb. Momentum built steadily. Two major spikes appeared in July and December. These aligned with mid-year promotions and the holiday shopping season. The line chart (bottom-right) compares monthly sales in 2020 (blue) and 2021 (orange). The contrast is clear. Sales in 2020 stayed flat and low. In 2021, they rose consistently, peaking in December. This surge was likely fueled by holiday demand and strong promotional efforts. The heatmap (center-left) supports this trend. It shows how monthly sales moved from dark red (low) to bright green (high) over the year. December 2021 posted the highest monthly sales at \$99,056,870. July followed with \$84,427,357. These were the two strongest months of the year. The top-right bar chart clearly compares total sales from 2020 and 2021. Sales jumped from \$109,561,561 in 2020 to \$790,340,564 in 2021.

That's a YoY growth rate of 621.37%. This growth wasn't just a recovery. It was a leap forward.

This performance shows a strong return in consumer demand after the pandemic. It also reflects Adidas' successful shift in strategy. The brand's ability to adapt and grow in a difficult market sets an important benchmark for recovery planning. Looking ahead, Adidas should continue to track detailed seasonal patterns. It should adjust inventory around peak periods and invest more in digital and promotional tools. The 2021 growth was likely driven by better e-commerce performance and targeted marketing. If scaled to other markets and supported by predictive analytics, these strategies could drive even more growth and improve efficiency.

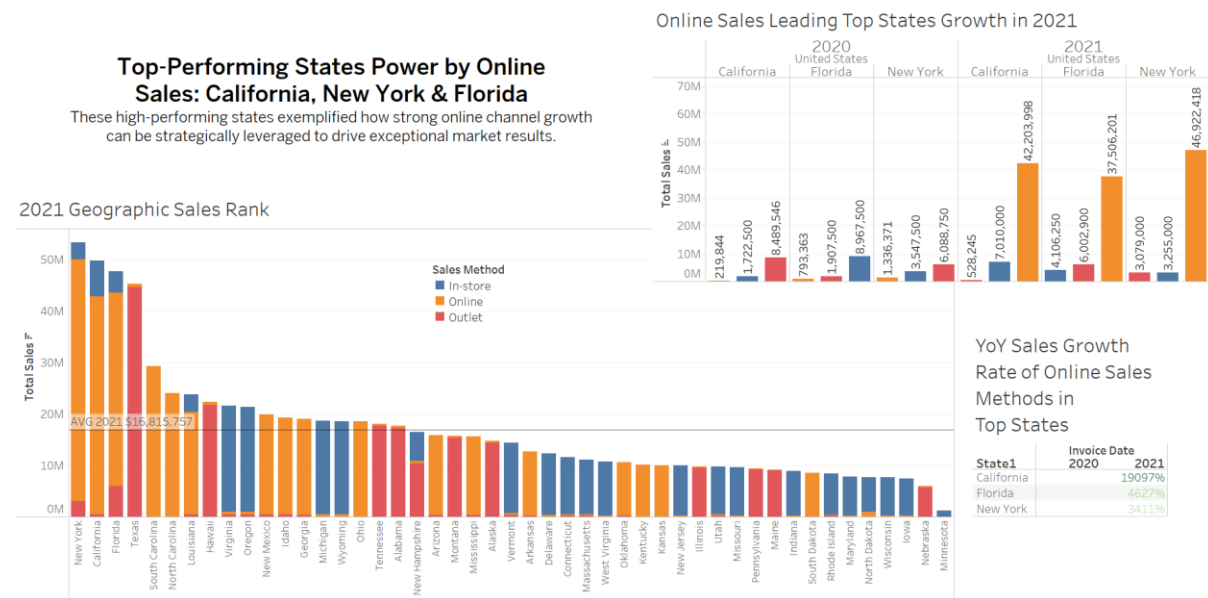
Dashboard 2



In 2021, Adidas' online sales channel in the U.S. experienced explosive growth. Sales increased by an incredible 7,425.08% year-over-year. This made online sales the main driver of total annual growth. It contributed heavily to both revenue and profit. Among all sales channels, online sales also recorded the highest average profit margin at 45.5%.

Sales trends showed that online performance started gaining momentum in the fourth quarter of 2020. By Q3 2021, online sales reached \$99,521,176. This figure was much higher than in-store sales (\$63,111,250) and outlet sales (\$54,857,179) during the same period. Physical stores did begin to recover after the pandemic. However, their growth was slower, and their profit margins remained lower compared to online sales. This contrast highlights a clear shift in consumer behavior toward digital shopping. It also shows that Adidas' investments in e-commerce, digital marketing, and agile supply chains were highly effective. Moving forward, Adidas should continue to build on its digital strategy. Key actions include expanding e-commerce platforms, improving user experience, and adopting smart inventory systems. A full integration of online and offline channels will also be critical.

Dashboard 3



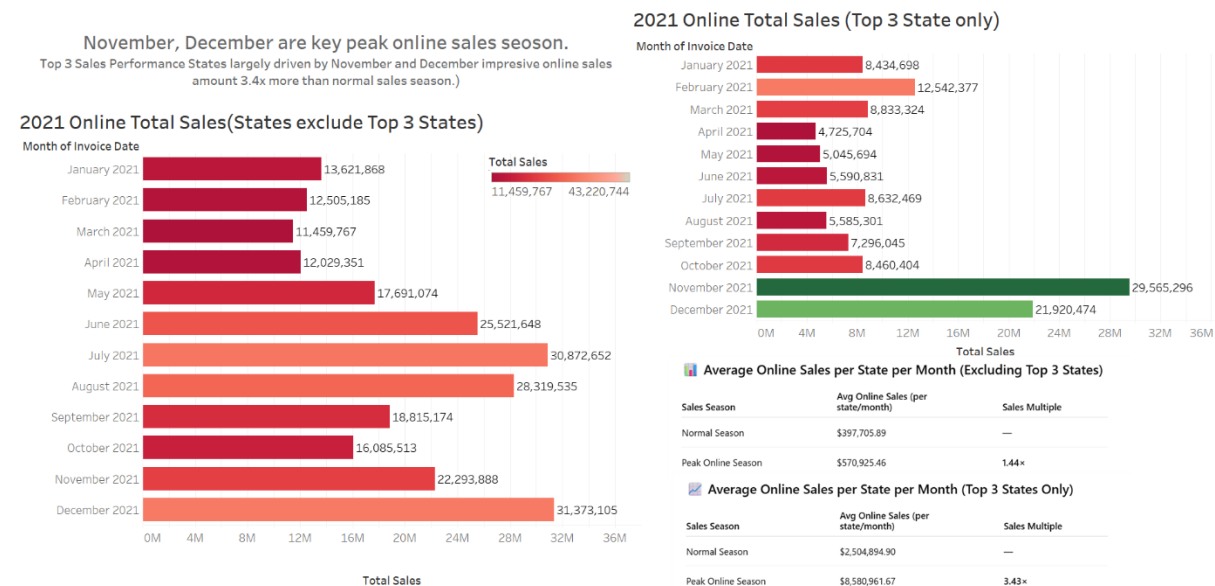
This dashboard highlights the key role of online sales in Adidas' regional growth strategy in the U.S. during 2021. California, Florida, and New York stood out as the strongest performers. Their year-over-year online sales growth was exceptional—California grew by 19,097.25%, Florida by 4,627.50%, and New York by 3,411.18%. These growth rates were far above the national average. This emphasizes the critical role these states played in Adidas' digital transformation.

In all three states, online channels outperformed both in-store and outlet sales. E-commerce contributed the largest share to total revenue. This firmly established online sales as the main engine of growth. The comparative chart clearly shows the jump in online sales from 2020 to 2021. In 2020, online growth was flat. But by 2021, there was a dramatic surge. This shift marks a clear change in consumer behavior—from physical shopping to digital platforms. This trend reflects several factors. It shows the impact of strong digital marketing and platform improvements. It also points to increased consumer reliance on online shopping during and after the pandemic. New York recorded the highest total sales among all U.S. states in 2021. California and Florida followed closely. All three exceeded the national average sales benchmark of \$16.8 million. These states have now become key markets for Adidas' online operations due to their size and growth potential.

Looking ahead, Adidas should treat California, Florida, and New York as high-priority digital hubs. The company should increase investments in local digital marketing. It should also expand warehouse and delivery capacity. In addition, Adidas should offer more online-only product options based on local consumer preferences. The digital strategies used in these states—such as mobile-first campaigns, targeted promotions, and easy checkout processes—worked well. Adidas can apply these proven tactics in other fast-growing markets. States like Texas, North Carolina, and Georgia are good candidates. These states show strong market potential and rising online engagement.

In conclusion, the success in California, Florida, and New York shows the growing importance of regional digital transformation. Adidas should continue investing in scalable e-commerce systems and data-driven marketing. This approach will strengthen its position in digital retail and unlock further growth across the U.S.

Dashboard 4



In dashboard 4 we drill-down analyse the 2021 online sales in monthly dimension. Bar chart of sales over time make sales trend clearly visible, the red-green diverging colour coded bar in chart make the significant high sales data of top 3 state in November and December clearly visible.

We break down the data to 2 visualisation chart which is 2021 monthly online total sales of top 3 state (California, Florida, New York) and 2021 monthly online total sales of other 47 states. Two chart shows difference of sales trend. The monthly sales of top 3 state show stable fluctuation from January to October, then explosively risen in November and December. Apart from that, other states remain stable fluctuation all the year.

According to this finding, we examine 2021 monthly online total sales of top 3 states and identify there are significant high sales in November and December. We define that November and December is peak online sales season, January to October is normal season. A table is there to shows the difference of monthly sales between peak online sales season and normal online sales season.

Besides, a sales multiplier of monthly peak online sales season compare normal online sales season will be the indicator of comparison between the performance improvement uplift of top 3 states and other states in peak online sales period. The sale multiplier of top 3 states is 3.43x high sales performance uplift, sales multiplier of other states is 1.44x only.

We can identify that top 3 states achieve huge performance uplift because they have well strategy to boosted online sales in peak online sales season, but other states don't have this strategy. Thereby other state missed up significant sales potential in November and December. If other states sales multiplier achieves at least 3.43x thereby we can define that the strategy boost peak online sales season sales has been effectively apply in other states too.

Dashboard 5

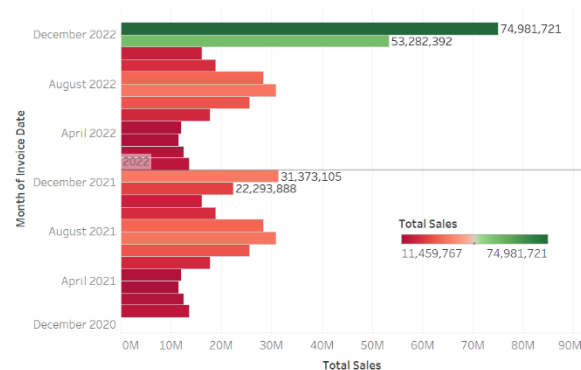
Applying the successful strategy from the top 3 states to other state results in a 9.44% increase in total sales.

Assuming 2022 sales remain flat compared to 2021, replicating the strategy of the top 3 states across other states would result in a 9.44% year-over-year increase in total sales for Adidas US.

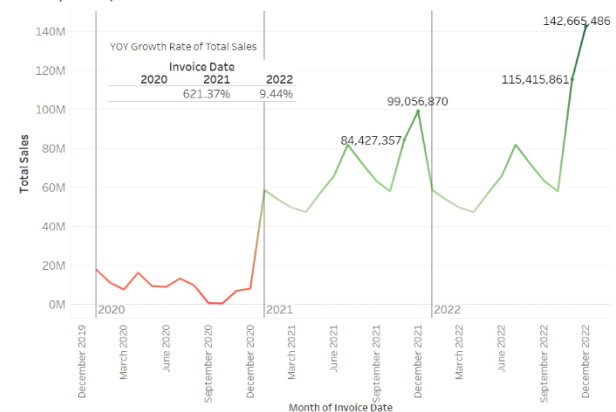
Average Online Sales per State per Month (Excluding Top 3 States)			Year ..
Sales Season	Avg Online Sales (per state/month)	Sales Multiple	2020
Normal Season	\$397,705.89	—	2021
Peak Online Season	\$1,327,394.68	3.34x	2022

2022 Online Total Sales (States exclude Top 3 States)

Invest more during the peak online sales season to increase monthly average sales from 1.44x to 3.43x (this multiple refers to average monthly sales in the peak season vs. the normal season — by replicating the success of the top 3 states and capturing strong demand).



2020,2021,2022 Total Sales Growth



Dashboard 5 is the show case of the impact of replicating the existing peak season online sales boost strategy which already achieve success sales uplift result in top 3 state and will be apply to all the other state.

For purpose of this analysis and to clearly demonstrate the positive impact, we assume that all 2022 sales data are identical to those of 2021. The only manipulated variable is the sales in peak online month of all state exclude top 3 states. In 2022, all Adidas US sales states are applied peak season online sales boost strategy and achieve success sales uplift result. all the state will achieve at least states sales multiplier 3.43x (monthly sales of peak online sales season compare normal online sales season)

From the line graph of 2020,2021,2022 sales, we can see the growth in November and December. Which is all positively impact only. Moreover, the data of total sales shows Adidas US sales growth from \$790,340,564(2021) to \$864,937,684(2022), which showed a year-over-year growth of 9.44%.

We can identify that if all states implement boost online sales strategy in peak online sales season will bring a 9.44% growth of Adidas US total sales.

Conclusion & Recommendation

An analysis of Adidas US online sales from 2020 to 2022 shows a clear trend. Online sales are the most profitable channel, with an average profit margin of 45.5%. California, Florida, and New York led the way in performance. These states saw strong growth during the peak online sales season in November and December. Their success came from effective digital marketing and focused online sales strategies.

However, this level of success was not consistent across the country. Many states failed to match the performance of the top three. The main reason was a lack of clear and unified online sales strategies. These states missed valuable opportunities during the most profitable time of the year. This gap shows the urgent need to apply proven strategies in all states. To close this gap and support long-term growth, one key strategy is recommended:

1. Strengthen online sales strategies across all states

High-performing states have already shown what works. Their online sales strategies should be applied nationwide. This includes using mobile-first website designs. It also means offering a simple and fast checkout process. In addition, each state should use digital marketing tailored to local customers.

2. Focus on growing high-margin online sales channels

Online sales offer higher profit margins than physical stores and outlets. This advantage has remained consistent over time. Adidas should invest more in this channel. A stronger online presence will directly support overall profit growth.

3. Boost advertising during the Christmas and year-end sales season

Customers tend to spend more during November and December. Adidas should increase marketing efforts during this peak period. In top-performing states, this approach led to a 3.43x sales increase. If used across all states, it could drive a 9.44% increase in annual sales.

4. Enhance online sales infrastructure

Adidas should improve its warehousing and delivery systems. The platform must also be scalable to manage high demand during peak seasons. Strong infrastructure helps maintain a consistent customer experience in every state.

5. Use promotions to encourage more customers to shop online

Adidas should run targeted campaigns in underperforming states. These can include online-only discounts, early access deals, and free shipping offers. Each promotion can help increase customer interest in online shopping.

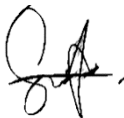
Adidas can take clear steps by following these data-driven recommendations. This approach will help the company strengthen its leadership in the online sportswear market. It will also boost sales across the country. In the long run, it will improve overall profitability.

Reference

1. Adidas_US_Sales_Datasets.xlsx

Heemali Chaudhari (2023). Adidas Sales Dataset: Adidas Sales in United States. Retrieve from: <https://www.kaggle.com/datasets/heemalichaudhari/adidas-sales-dataset?resource=download>

Team Agreement

TEAM MEMBERS	CONTACT DETAILS	
	MOBILE	EMAIL
Han Bin	89613643	hanbin02202004@gmail.com
Chong Shi Xian	84059825	shixianck07@gmail.com
INFORMAL COMMUNICATION - <i>We have decided</i> 1) To be respectful when communicating. 2) To be open to feedback from each other. 3) To respond to each other's messages in a timely manner.		
MEETINGS - <i>We have decided</i> 1) To come prepared and ready to contribute. 2) To actively listen to each other. 3) To follow the agenda and stay on track.		
MAKING DECISIONS - <i>We have agreed</i> 1) To involve all team members in decision making. 2) To consider all options before making a decision. 3) To discuss and debate different viewpoints. 4) Will strive for consensus whenever possible. 5) Will document all decisions and take actions.		
SANCTIONS We hope to work in harmony together. We have different strengths. We accept that this is a group piece of work and we are all responsible for doing our best. However, we agree now that *If individuals have difficulties in working with the team or on the task, we will try to sort them out promptly by talking with each other. *We will seek advice - as soon as is possible - from our tutor for those serious problems which we cannot resolve ourselves.		
SIGED  		

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in your completion of:
MODULE NAME: **Academic Integrity**

on this date: **Wednesday, 10 July 2024**

The completion of this course indicates that you have familiarised yourself with the principles and requirements of good academic practice when researching and submitting coursework.

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in your completion of:
MODULE NAME: **Academic Integrity**

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Individual Papers

Name: Han Bin (24285587)

1. The project promotes course learning

I have a more systematic understanding of the entire process from raw data collection to visual presentation. I have mastered the skills of using Tableau to display charts and dashboards, and Excel data preprocessing.

2. Group cooperation experience

The teamwork atmosphere is good and everyone helps each other. I am mainly responsible for sales trends and regional analysis, and also assist in establishing data visualization charts and formatting

3. Project deficiencies and challenges

Although the final result basically achieved the goal, there were also some challenges in the project: the initial draft of the visualization plan lacked coherence and professionalism, and it took several rounds of adjustments to make the chart more logical and professional.

Name: Chong Shi Xian (24285579)

1. Gains from the course

We conducted a multi-dimensional analysis of Adidas' sales data in the United States, including sales trends, product category performance, regional distribution, etc., and used data visualization to help identify market trends. It strengthened my ability to analyze data

2. Teamwork performance

Team members actively cooperated with each other and supported each other when encountering problems. In the group, I was mainly responsible for the performance of sales methods in various channels, the analysis of sales peak performance, and assisted in optimizing visualization charts

3. Problems in the project

In the early stage of the project, I was not proficient in data visualization and failed to solve the business problem. Later, after more attempts, I finally achieved the goal and successfully summarized the analysis results